



2024 PRELIMINARY YEARLY EXAMINATION

Mathematics Extension

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I – 10 marks (pages 2 – 4)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 5 – 8)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer page in the writing booklet for Questions 1 – 10.

- 1** How many ways can 7 people be seated around a circular table if 2 particular people, Preksha and Chloe, do not wish to sit next to each other?

A. 480
B. 718
C. 2880
D. 3360

- 2** A parabola has the parametric equations $x = t - 2$ and $y = 2t^2 - 1$. What is the Cartesian equation of the parabola?

A. $y = 2x^2 + 4x - 1$
B. $y = 2x^2 + 4x + 1$
C. $y = 2x^2 + 8x - 7$
D. $y = 2x^2 + 8x + 7$

- 3** When $P(x)$ is divided by $x^2 + 5x - 6$, the remainder is the polynomial $R(x) = 2x - 5$. What is the remainder when $P(x)$ is divided by $(x - 1)$?

A. -7
B. -6
C. -3
D. 7

- 4 If $\frac{\pi}{2} < A < \pi$, and $\pi < B < \frac{3\pi}{2}$ with $\cos A = t$ and $\sin B = t$, then which of the following is equivalent to $\sin(A + B)$?

A. 0
B. 1
C. $2t^2 - 1$
D. $1 - 2t^2$

- 5 Which of the following is equivalent to $\frac{1 - \cos 2x}{\sin 2x}$?

A. $1 - \cot 2x$
B. 1
C. $\cot x$
D. $\tan x$

- 6 Which of the following is an expression for $\frac{dy}{dt}$ if $y = \sqrt{25 - x^2}$ and $\frac{dx}{dt} = 2$?

A. $-4x\sqrt{25 - x^2}$
B. $-2x\sqrt{25 - x^2}$
C. $\frac{-4x}{\sqrt{25 - x^2}}$
D. $\frac{-2x}{\sqrt{25 - x^2}}$

- 7 The solutions to the equation $\sin 11\theta \sin 3\theta = \alpha$, will always be the same as the solutions to which of the following equations?
- A. $\cos 14\theta - \cos 8\theta = \alpha$
 - B. $\cos 8\theta - \cos 14\theta = \alpha$
 - C. $\cos 14\theta - \cos 8\theta = 2\alpha$
 - D. $\cos 8\theta - \cos 14\theta = 2\alpha$
-
- 8 Let $(x-1)P(x) = 16x^5 - 20x^3 + 5x - 1$. If $P(x) = (4x^2 + ax - 1)^2$, what is the value of a ?
- A. 0
 - B. $\frac{1}{2}$
 - C. 1
 - D. 2
-
- 9 Lawrence is about to walk up a set of stairs with 8 steps. He can take either one or two steps at a time. In how many different ways can he walk up the set of stairs?
- A. 5
 - B. 20
 - C. 34
 - D. 163
-
- 10 In how many ways can the letters of the word **COMPLEMENT** be arranged if the letter **M**'s are separated and the letter **E**'s are separated?
- A. 544320
 - B. 545580
 - C. 564480
 - D. 584640

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate section of the writing booklet. Extra writing paper is available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

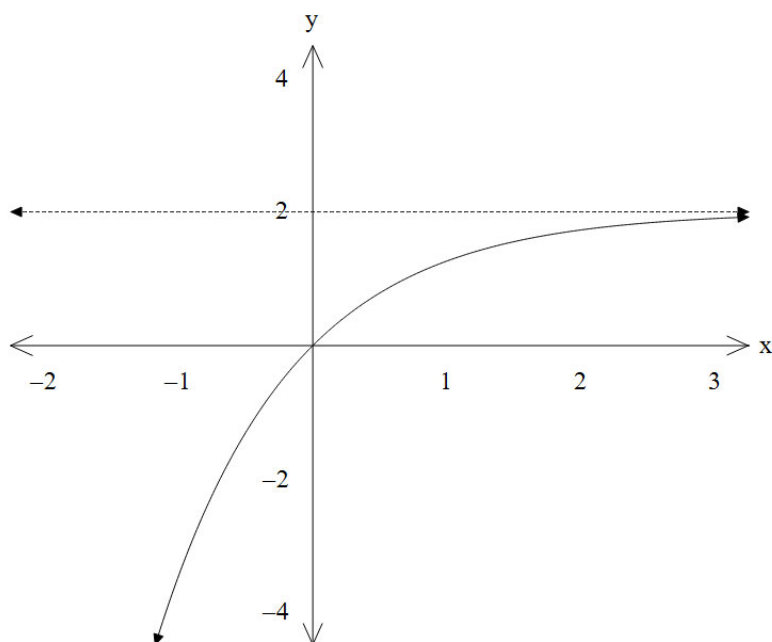
Question 11 (16 marks) Use the Question 11 section of the writing booklet.

- (a) Use a compound angle formula to find the exact value of $\cos 75^\circ$. 2
- (b) (i) Find the value of k if $x - 2$ is a factor of $P(x) = 2x^3 + x^2 - kx + 6$. 1
- (ii) Hence, solve $P(x) = 0$. 2
- (c) (i) Show that $\cot \theta - \cot 2\theta = \operatorname{cosec} 2\theta$. 2
- (ii) Hence find the exact value of $\cot 15^\circ$. 1
- (d) If α, β and γ are roots of $2x^3 - 5x^2 + 3x - 5 = 0$, find the value of:
- (i) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
- (ii) $\frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\beta}$. 2
- (e) Solve $\frac{x+4}{x-3} \leq 2$. 3
- (f) The **digital sum** of a natural number is defined to be the sum of its digits. For example, the digital sum of 123 is $1 + 2 + 3 = 6$.
- Nineteen two-digit numbers are randomly selected. Prove that at least two of them have the same digital sum. 2

End of Question 11

Question 12 (16 marks) Use the Question 12 section of the writing booklet.

(a) The diagram shows the graph of $f(x) = 2 - 2e^{-x}$.



On the separate grids provided in your answer booklet, sketch the graphs of:

- | | |
|--|---|
| (i) $y = \frac{1}{f(x)}$ | 2 |
| (ii) $ y = f(-x)$ | 2 |
| | |
| (b) (i) Find the domain AND the range of the function $y = \sin^{-1}\left(\frac{x}{4} - 1\right)$. | 2 |
| (ii) Sketch the graph of the function $y = \sin^{-1}\left(\frac{x}{4} - 1\right)$. | 2 |
| | |
| (c) A polynomial is given by $P(x) = x^5 - 2x^4 - 6x^3 + 20x^2 - 19x + 6$. | |
| (i) Prove that $x = 1$ is a root of $P(x)$ of multiplicity 3. | 2 |
| (ii) Factorise $P(x)$ completely. | 2 |
| (iii) Hence, solve $P(x) \geq 0$. | 1 |
| | |
| (d) Prove that $\sin\left(2\sin^{-1}x + \cos^{-1}x\right) = \sqrt{1-x^2}$. | 3 |

Question 13 (15 marks) Use the Question 13 section of the writing booklet.

- (a) An egg at room temperature, 20°C , is placed in a saucepan of boiling water which is maintained at 100°C . When the egg has been in the boiling water for t minutes the internal temperature of the egg is $T^{\circ}\text{C}$. The rate at which the internal temperature of the egg rises is proportional to the difference between the egg's internal temperature and that of the boiling water, so that T satisfies the equation $\frac{dT}{dt} = k(T - 100)$, where k is a constant.

(i) Show that $T = 100 + Ae^{kt}$ satisfies the equation. 1

(ii) The internal temperature of the egg rises to 60°C after 10 minutes.
Find the values of A and k . 2

(iii) How long does it take the internal temperature of the egg to reach 90°C ? 1

(iv) What would happen to the internal temperature of the egg if the egg was left in the water indefinitely? Justify your answer. 1

- (b) A function is defined by $f(x) = (x + 2)^2 - 9$, for $-2 \leq x \leq 2$.

(i) Find the equation of the inverse function $y = f^{-1}(x)$. 1

(ii) On the same number plane, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing clearly the coordinates of the intercepts on the coordinate axes and the endpoints. 3

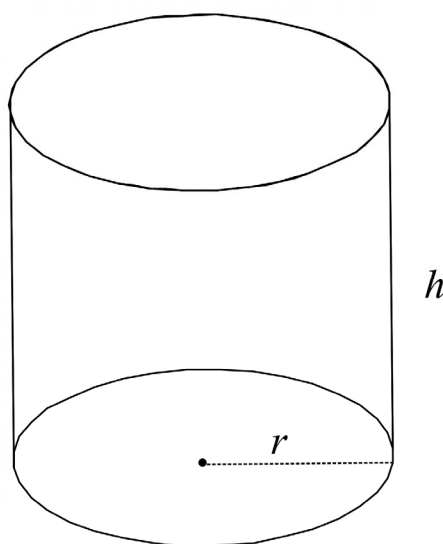
(iii) Find the x -coordinate of the point of intersection of the curves $y = f(x)$ and $y = f^{-1}(x)$, giving the answer in simplest exact form. 2

- (c) (i) Write down the expansion of $x(1+x)^n$ in ascending powers of x . 1

(ii) Hence show that $2^n C_1 + 3^n C_2 + \dots + n^n C_{n-1} = (n+2)(2^{n-1} - 1)$. 3

Question 14 (13 marks) Use the Question 14 section of the writing booklet.

- (a) A balloon in the shape of a cylinder with height h and radius r , expands so that h is always proportional to r , that is $h = kr$ for some constant k . When $r = 4$ cm. the volume is increasing at the rate of $0.2 \text{ cm}^3\text{s}^{-1}$.
 [The formula for the volume of a cylinder is $V = \pi r^2 h$ and the formula for the surface area of a cylinder is $A = 2\pi r(r + h)$]

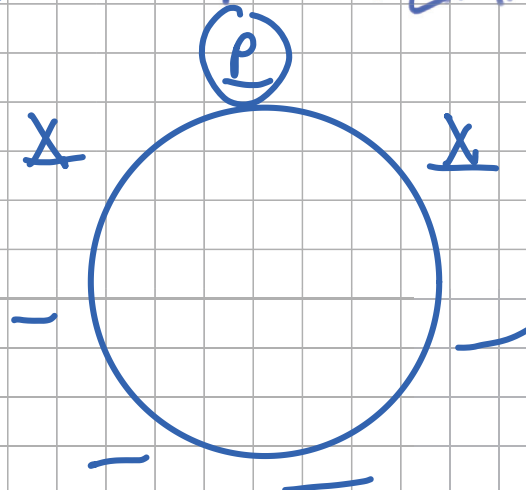


- (i) Show that when $r = 4$ cm, the rate of change of the radius is given by 2

$$\frac{dr}{dt} = \frac{1}{240\pi k} \text{ cms}^{-1}.$$
- (ii) If the surface area of the balloon is increasing at the rate of $0.1 \text{ cm}^2\text{s}^{-1}$ when $r = 4$ cm, find the value of the constant of proportionality, k . 3
- (b) Consider the word **PHOTOGRAPHS**.
- (i) How many arrangements of the letters of the word are possible? 2
- (ii) What is the probability that there are exactly four letters between the two **H**'s when the letters of the word **PHOTOGRAPHS** are arranged in a straight line? 2
- (c) (i) Show that ${}^{n+1}C_r - {}^nC_r = {}^nC_{r-1}$, for $r = 1, 2, 3, \dots, n$. 2
- (ii) Hence find the value of ${}^3C_2 + {}^4C_2 + {}^5C_2 + \dots + {}^{100}C_2$. 2

End of Paper

XI 2024 PRELIMINARY EXAM SOLUTIONS

1.  sit Preksha in 1 way
sit Chloë in 4 ways
remaining people in $5!$ ways
 $1 \times 4 \times 5!$
 $= 480$

$\therefore$ (A)

2. $\lambda = t - 2$

$t = \lambda + 2$

$$y = 2t^2 - 1$$
$$= 2(\lambda + 2)^2 - 1$$

$$y = 2\lambda^2 + 8\lambda + 7$$

$\therefore$ (D)

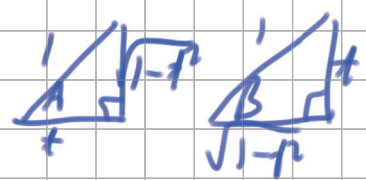
3. $P(x) = (x^2 + 5x - 6)Q(x) + (2x - 5)$

$$P(1) = 0 \cdot Q(1) + 2 - 5$$

$$P(1) = -3$$

$\therefore$ (C)

$$\begin{aligned}
 4. \quad \sin(A+B) &= \sin A \cos B + \cos A \sin B \\
 &= \sqrt{1-t^2}(-\sqrt{1-t^2}) + t \times 1 \\
 &= -(1-t^2) + t^2 \\
 &= -1 + 2t^2 \\
 \therefore \textcircled{C}
 \end{aligned}$$

$\cos A = t \quad \sin B = t$

 $\sin A = \sqrt{1-t^2} \quad \sin B = t$
 $\cos A = t \quad \cos B = -\sqrt{1-t^2}$

$$\begin{aligned}
 5. \quad \frac{1 - \cos 2x}{\sin 2x} &= \frac{2\sin^2 x}{2\sin x \cos x} \\
 &= \tan x \\
 \therefore \textcircled{D}
 \end{aligned}$$

$$\begin{aligned}
 \cos 2x &= 1 - 2\sin^2 x \\
 \sin 2x &= 2\sin x \cos x
 \end{aligned}$$

$$\begin{aligned}
 6. \quad \frac{dy}{dt} &= \frac{dy}{dx} \times \frac{dx}{dt} \\
 &= \frac{1}{2}(25-x^2)^{-\frac{1}{2}}(-2x) \times 2 \\
 &= \frac{-2x}{\sqrt{25-x^2}} \quad \therefore \textcircled{D}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \sin 11\theta \sin 3\theta &= \alpha \\
 \frac{1}{2}(\cos(11\theta - 3\theta) - \cos(11\theta + 3\theta)) &= \alpha \\
 \cos 8\theta - \cos 14\theta &= 2\alpha \\
 \therefore \textcircled{D}
 \end{aligned}$$

$$8. (x-1)(4x^2+ax-1)^2 = 16x^5 - 20x^3 + 5x - 1$$

$$\text{when } x = \frac{1}{2} \quad \left(-\frac{1}{2}\right)\left(1 + \frac{a}{2} - 1\right)^2 = 16 \times \frac{1}{32} - \frac{20}{8} + \frac{5}{2} - 1$$

$$-\frac{a^2}{8} = \frac{1}{2} - \frac{5}{2} + \frac{5}{2} - 1$$

$$-\frac{a^2}{8} = -\frac{1}{2}$$

$$a^2 = 4$$

$$a = \pm 2$$

$$\text{when } a = 2, (16 + 12a - 1)^2 = 361$$

$$2a + 15 = \pm 19$$

$$2a = 4 \quad \text{or} \quad 2a = -34$$

$$a = 2 \quad \text{or} \quad a = -17$$

$$\therefore a = 2 \therefore \textcircled{D}$$

9. Double steps | Single steps

$$0 \quad | \quad 8 \quad SSSSSSSS = 1 \text{ way}$$

$$1 \quad | \quad 6 \quad OSSSSS = 7 \text{ ways}$$

$$2 \quad | \quad 4 \quad OOSSS = \frac{6!}{4!2!} = 15 \text{ ways}$$

$$3 \quad | \quad 2 \quad OOOSS = \frac{5!}{3!2!} = 10 \text{ ways}$$

$$4 \quad | \quad 0 \quad OOOOO = 1 \text{ way}$$

$$\text{Total} = 1 + 7 + 15 + 10 + 1$$

$$= 34 \text{ ways}$$

$$\therefore \textcircled{C}$$

10. C O M P L E N T
M E

Total - M's and E's together - M's together (E not together)
- E's together (M's not together)

ms and E's together

mm EE COMPLNT

ways = $8!$

m_i together ES separate

-(NM)-C-O-P-L-N-

$$\# \text{ ways} = 7! \times {}^6C_2 = 141120$$

similarly for E_i together, M_i separate

ways = 141120

$$\therefore \text{Possibilities} = \frac{10!}{2!2!} - 8! - 2 \times 141120$$
$$= 584640$$

0

Alternatively,

of ways = total - E's together - M's together + E's and M's together

$$= \frac{10!}{2!2!} - \left(\frac{9!}{2}\right) \times 2 + 8!$$

COMPLEMENT

$= 584640$

$$\begin{aligned}
 11. a) \cos 75^\circ &= \cos(45+30)^\circ \\
 &= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\
 &= \frac{1}{\sqrt{2}} \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \frac{1}{2} \\
 &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \\
 &= \frac{\sqrt{3}-1}{2\sqrt{2}}
 \end{aligned}$$

② correct solution

① correct expansion of $\cos(45+30)^\circ$ or equivalent

$$\begin{aligned}
 b) i) P(2) &= 0 \\
 0 &= 16 + 4 - 2k + 6 \\
 0 &= 26 - 2k \\
 2k &= 26 \\
 k &= 13
 \end{aligned}$$

① correct answer

$$\begin{aligned}
 b) ii) 2x^3 + x^2 - 13x + 6 &= 0 \\
 (x-2)(2x^2 + 5x - 3) &= 0 \\
 (x-2)(2x-1)(x+3) &= 0 \\
 \therefore x &= -3, \frac{1}{2}, 2
 \end{aligned}$$

② correct solution

① factorises $P(x)$ into two terms

or ① uses sum of product of roots and attempts to solve simultaneously

or let roots be $\alpha, \beta, 2$.

$$\sum \alpha: \alpha + \beta + 2 = -\frac{1}{2}$$

$$\sum \alpha\beta: 2\alpha\beta = -3$$

$$\alpha\beta = -\frac{3}{2}$$

$$\alpha + \frac{-3}{2\alpha} = -\frac{1}{2}$$

$$2x^2 - 3 = -5x$$

$$2x^2 + 5x - 3 = 0$$

$$(2x - 1)(x + 3) = 0$$

$$x = \frac{1}{2} \text{ or } x = -3$$

$$\therefore x = -3, \frac{1}{2} \text{ and } 2$$

$$11 \text{ c) i) } \cot \theta - \cot 2\theta$$

$$= \frac{\cos \theta}{\sin \theta} - \frac{\cos 2\theta}{\sin 2\theta}$$

$$= \frac{2 \cos^2 \theta}{2 \sin \theta \cos \theta} - \frac{\cos 2\theta}{\sin 2\theta}$$

$$= \frac{2 \cos^2 \theta - \cos 2\theta}{\sin 2\theta} \quad (\sin 2\theta = 2 \sin \theta \cos \theta)$$

$$= \frac{1}{\sin 2\theta} \quad (\sin 2\theta \cos 2\theta = 2 \cos^2 \theta - 1)$$

$$= \operatorname{cosec} 2\theta$$

$$\text{OR let } t = \tan \frac{\theta}{2}$$

$$\begin{aligned} \cot \theta - \cot 2\theta &= \frac{1}{t} - \frac{1-t^2}{2t} \\ &= \frac{2 - (1-t^2)}{2t} \\ &= \frac{1+t^2}{2t} \\ &= \operatorname{cosec} 2\theta \end{aligned}$$

$$\text{ii) } \cot \theta - \cot 2\theta = \operatorname{cosec} 2\theta$$

$$\text{let } \theta = 15^\circ$$

$$\cot 15^\circ = \operatorname{cosec} 30^\circ + \cot 30^\circ$$

$$= 2 + \sqrt{3}$$

② correct proof

① uses a double angle result correctly

OR ① expresses $\cot \theta$ and $\cot 2\theta$ in terms of t results

($\sin 2\theta = 2 \sin \theta \cos \theta$)

① correct answer (must use ii))

$$11d) i) \alpha\beta + \alpha\gamma + \beta\gamma = \frac{3}{2}$$

$$\begin{aligned} ii) & \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\beta} \\ &= \frac{\beta}{\alpha\beta\gamma} + \frac{\alpha}{\alpha\beta\gamma} + \frac{\gamma}{\alpha\beta\gamma} \\ &= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} \\ &= \frac{\frac{5}{2}}{\frac{5}{2}} \\ &= 1 \end{aligned}$$

① correct answer

② correct solution

① finds $\alpha + \beta + \gamma$ or $\alpha\beta\gamma$

$$11e) \frac{x+4}{x-3} \leq 2 \quad x \neq 3$$

$$x+4 = 2(x-3)$$

$$x+4 = 2x-6$$

$$x = 10$$



$$x < 3 \text{ or } x \geq 10$$

③ correct solution

② finds both critical values

① finds $x=10$ as a critical value

or ① multiplies both sides by $(x-3)^2$

11 f)

10	1
11	2
$\vdots$	$\vdots$
99	18

Digital sums

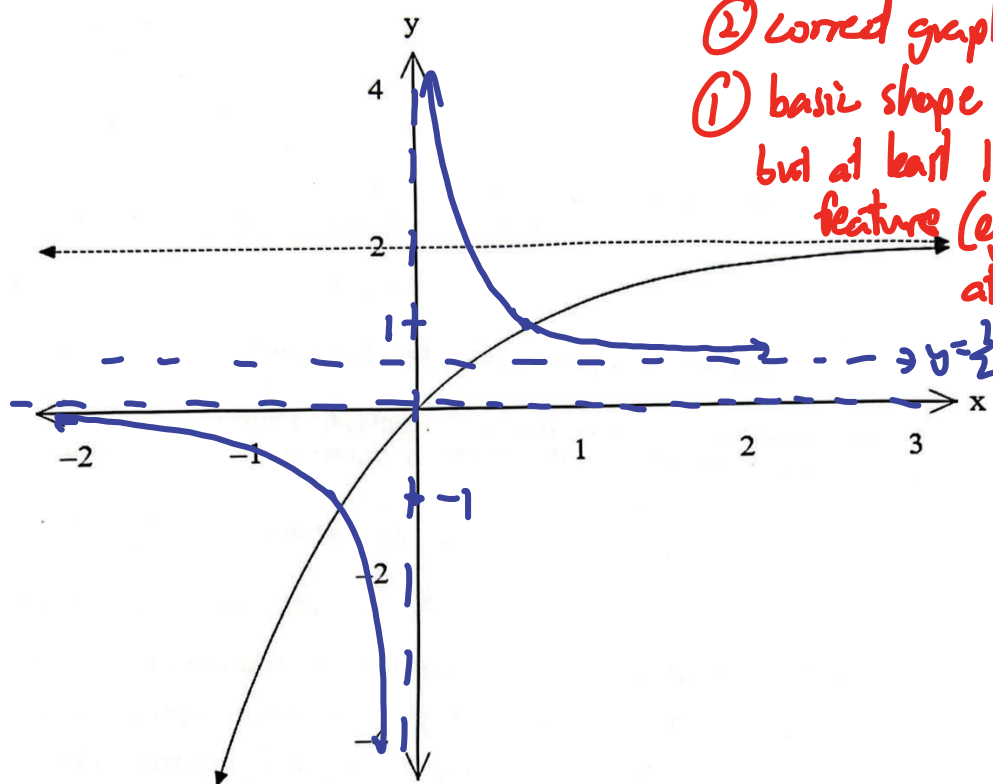
② correct solution

① identifies possible digital sums

Possible digital sums = 18

since 19 two digit numbers are chosen, by the pigeon hole principle there must be at least two of them that have the same digital sum

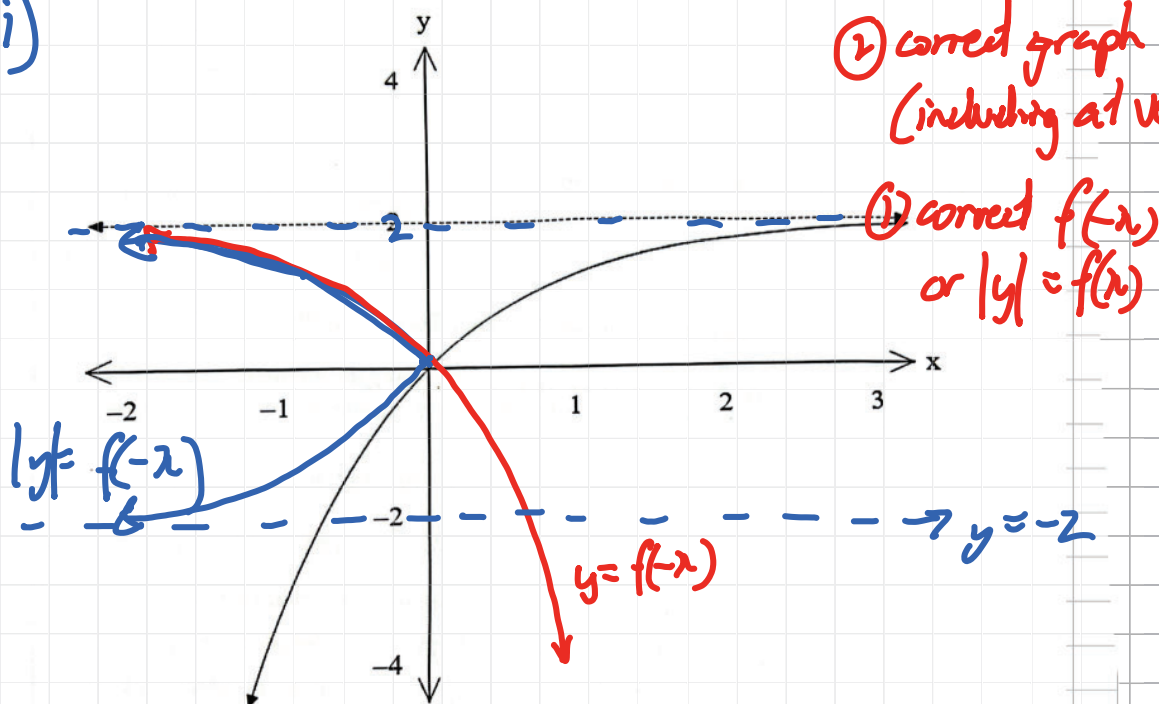
12 a i)



② correct graph

① basic shape correct
but at least 1 missing
feature (eg asymptote
at $y = \frac{1}{2}$)

12 a(ii)



② correct graph (including all vertices)

① correct $f(x)$ or $|y| = f(x)$

12 b) i) $D: -1 \leq \frac{x}{4} \leq 1$

$0 \leq \frac{x}{4} \leq 2$

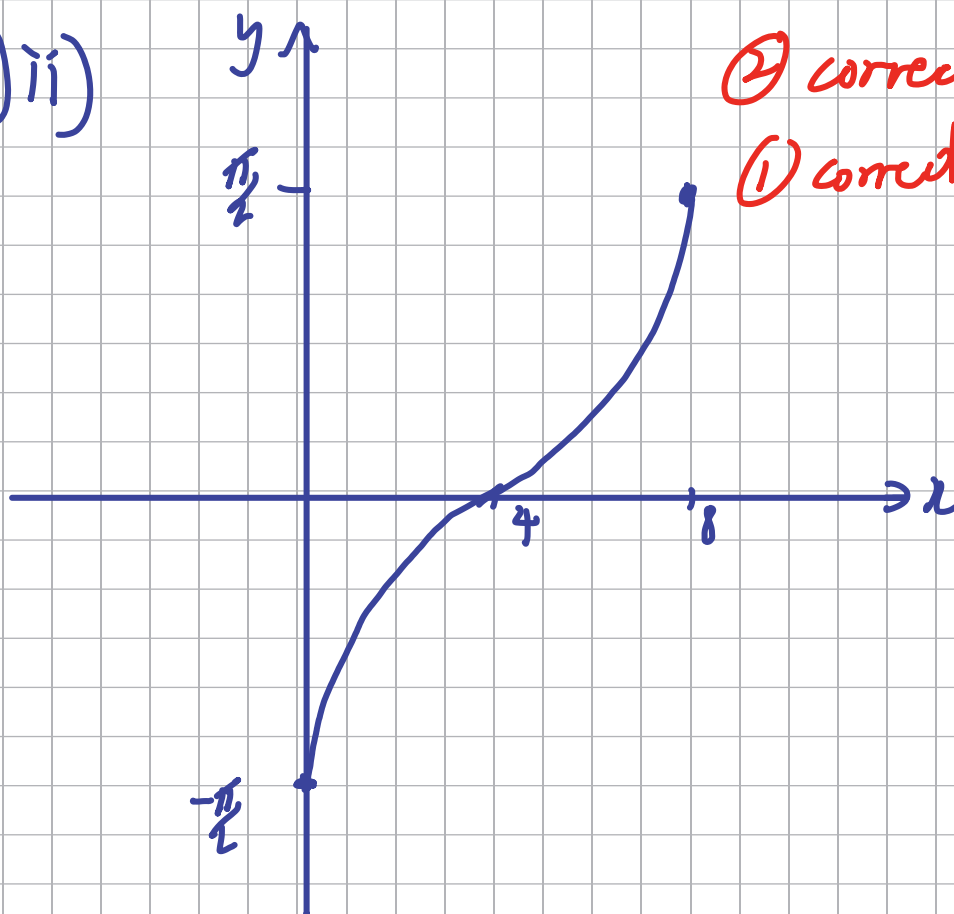
$0 \leq x \leq 8$

$R: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

② correct domain and range

① correct domain or range

12 b) ii)



② correct graph

① correct basic shape

$$12 \text{ C i)} \quad P(x) = x^5 - 2x^4 - 6x^3 + 20x^2 - 19x + 6$$

$$P'(x) = 5x^4 - 8x^3 - 18x^2 + 40x - 19$$

$$P''(x) = 20x^3 - 24x^2 - 36x + 40$$

$$P'''(x) = 60x^2 - 48x - 36$$

$$P(1) = 1 - 2 - 6 + 20 - 19 + 6$$

$$P'(1) \stackrel{=0}{=} 5 - 8 - 18 + 40 - 19$$

$$P''(1) \stackrel{=0}{=} 20 - 24 - 36 + 40$$

$$P'''(1) \stackrel{=0}{=} -24 \therefore \text{not multiplicity 4.}$$

$\therefore$ Root of multiplicity 3.

$$12 \text{ C ii)} \quad P(x) = (x-1)^3(x-\alpha)(x-\beta)$$

$$\text{Sum of roots } 1+1+1+\alpha+\beta = 2$$

$$\alpha+\beta = -1$$

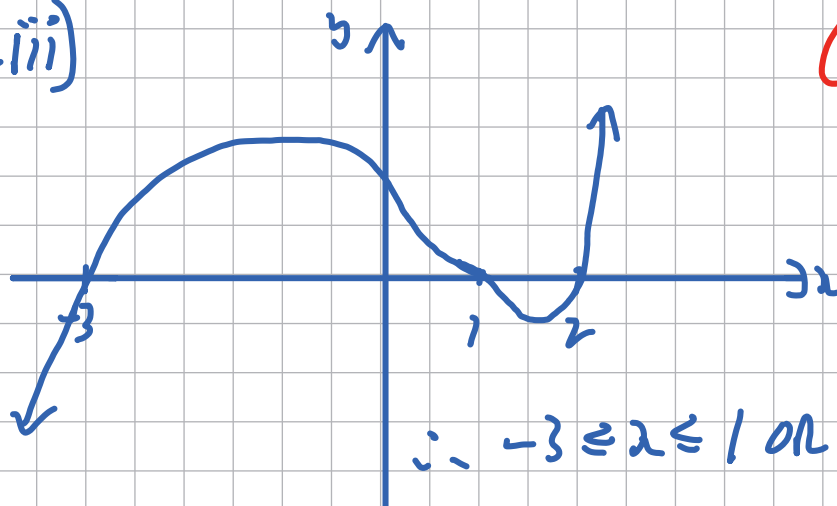
$$\text{Product of roots } 1 \times 1 \times 1 \times \alpha \times \beta = -6$$

$$\alpha\beta = -6$$

$$\alpha = -3, \beta = 2$$

$$\therefore P(x) = (x-1)^3(x+3)(x-2)$$

12 C iii)



$$\therefore -3 \leq x \leq 1 \text{ or } x \geq 2$$

② correct solution

① shows $P(1), P'(1), P''(1)$ and $P'''(1)$ are 0.

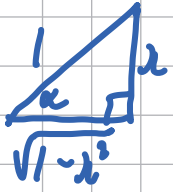
② correct answer

① obtains x^2+x-6 as a factor by long division or inspection

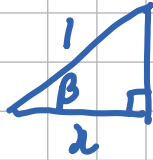
or ① obtains $\alpha+\beta=-1$ and $\alpha\beta=-6$

① correct answer

12d let $\alpha = \sin^{-1} x$, $\beta = \cos^{-1} x$



$\sin \alpha = x$



$\cos \beta = x$

$$\sin(2\sin^{-1}x + \cos^{-1}x)$$

$$= \sin(2\alpha + \beta)$$

$$= \sin 2\alpha \cos \beta + \cos 2\alpha \sin \beta$$

$$= 2 \sin \alpha \cos \alpha \cos \beta + (1 - 2 \sin^2 \alpha) \sin \beta$$

$$= 2x^2 \sqrt{1-x^2} + (1 - 2x^2) \sqrt{1-x^2}$$

$$= \sqrt{1-x^2} (2x^2 + 1 - 2x^2)$$

$$= \sqrt{1-x^2} \text{ as req'd.}$$

13a) i) $T = 100 + Ae^{kt}$

$$\frac{dT}{dt} = kAe^{kt}$$

$$= k(100 + Ae^{kt} - 100)$$

$$= k(T - 100) \quad (\text{using } T = 100 + Ae^{kt})$$

ii) When $t = 0$, $T = 20$

$$20 = 100 + Ae^0$$

$$A = -80$$

$$\therefore T = 100 - 80e^{kt}$$

When $t = 10$, $T = 60$

$$60 = 100 - 80e^{10k}$$

$$80e^{10k} = 40$$

$$e^{10k} = \frac{1}{2}$$

$$10k = \ln\left(\frac{1}{2}\right)$$

$$k = \frac{-\ln 2}{10}$$

$$\therefore A = -80 \text{ and } k = \frac{-\ln 2}{10}$$

③ correct solution

④ uses correct expression for expansion of $\sin(2\alpha + \beta)$ or equivalent and expresses one of the terms in x

(ie $\cos \alpha = \sqrt{1-x^2}$

or $\sin \beta = \sqrt{1-x^2}$

① writes correct expression for $\sin(2\alpha + \beta)$

OR finds $\cos \alpha = \sqrt{1-x^2}$

or $\sin \beta = \sqrt{1-x^2}$

① correct proof

② correct solution

① finds value of A .

ii) When $T = 90$

$$90 = 100 - 80e^{kt}$$

$$80e^{kt} = 10$$

$$e^{kt} = \frac{1}{8}$$

$$kt = \ln\left(\frac{1}{8}\right)$$

$$t = \frac{-10}{\ln 2} \times -\ln 8$$

$$t = 30$$

$\therefore$ Takes 30 minutes for internal temperature to reach 90°C .

iv) $T = 100 - 80e^{kt}$

As $t \rightarrow \infty$, $e^{-\frac{1}{10}\ln 2 t} \rightarrow 0$

$$\therefore T \rightarrow 100 - 80 \times 0 = 100$$

$\therefore$ Internal temperature approaches 100°C

① correct answer

① correct answer
(with justification)

13 b) i) $f^{-1}(x): x = (y+2)^2 - 9$

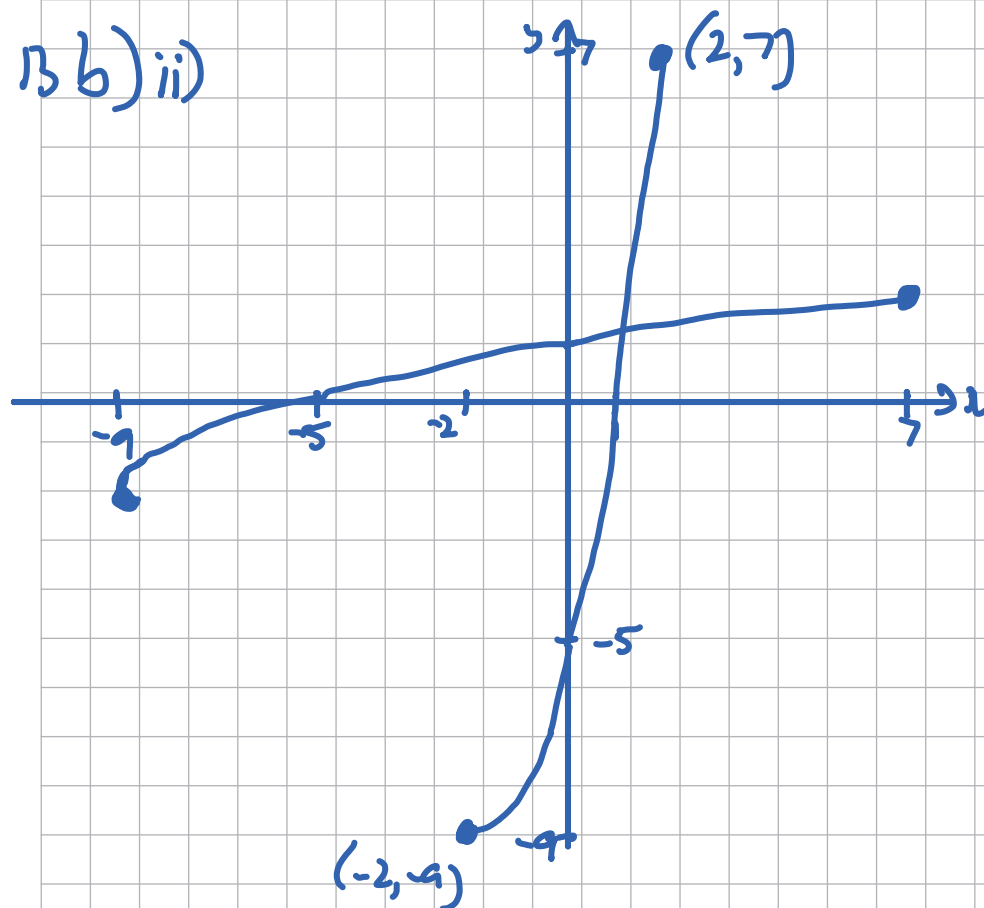
$-2 \leq y \leq 2$ ① correct equation

$$(y+2)^2 = x+9$$

$$y+2 = \pm \sqrt{x+9}$$

$$f^{-1}(x) = -2 + \sqrt{x+9}$$

B6) ii)



③ Correct graphs with intercepts and end-points (including shape)

② are correct graph with endpoints and intercepts

② Both graphs correct but missing or incorrect endpoints and/or intercepts

① one correct graph with missing and/or incorrect endpoints or intercepts

B6) iii)

$$x = (x+2)^2 - 9$$

$$x^2 + 4x - 5 = x$$

$$x^2 + 3x - 5 = 0$$

$$x = \frac{-3 \pm \sqrt{9+20}}{2}$$

$$x = \frac{-3 \pm \sqrt{29}}{2}$$

But $-2 \leq x \leq 2$, $\therefore x = \frac{\sqrt{29} - 3}{2}$

or $x = -2 + \sqrt{x+9}$
 $x+2 = \sqrt{x+9}$

$$x^2 + 4x + 4 = x + 9$$

$$x^2 + 3x - 5 = 0$$

$$x = \frac{-3 \pm \sqrt{29}}{2}$$

② correct solution

① obtains two x values but doesn't eliminate the incorrect solution

① correct expansion

$$13 \text{ c i) } x(1+x)^n = x \left[\binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n \right] \\ = x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \dots + \binom{n}{n-1}x^n + \binom{n}{n}x^{n+1}$$

ii) Differentiate w.r.t x

③ correct proof

② differentiates both sides & substitutes $x=1$

① differentiates one side and substitutes $x=1$

$$(1+x)^n \cdot 1 + x \cdot n(1+x)^{n-1} = 1 + 2^n \binom{n}{1}x + 3^n \binom{n}{2}x^2 + \dots + n \binom{n}{n-1}x^{n-1} + (n+1)x^n$$

Let $x=1$

$$2^n + n \cdot 2^{n-1} = 1 + 2^n \binom{n}{1} + 3^n \binom{n}{2} + \dots + n \binom{n}{n-1} + (n+1)$$

$$\therefore 2^n \binom{n}{1} + 3^n \binom{n}{2} + \dots + n \binom{n}{n-1} = 2^n + n \cdot 2^{n-1} - 1 - n - 1$$

$$= 2^n + n \cdot 2^{n-1} - (n+2)$$

$$= 2^{n-1}(2+n) - 1(n+2)$$

$$= (n+2)(2^{n-1} - 1)$$

14 a) i)



$$h = kr$$

② correct proof

① obtains $\frac{dV}{dr}$

$$V = \pi r^2 h$$

$$\text{But } h = kr$$

$$\therefore V = \pi k r^3$$

$$\frac{dV}{dr} = 3\pi k r^2$$

$$\frac{dV}{dr} = 4 \text{ where } r=4$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$\frac{dV}{dt} = 3\pi k r^2 \times \frac{dr}{dt}$$

when $r=4$

$$\frac{1}{5} = 48\pi k \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{240\pi k}$$

$$\begin{aligned} \text{ii) } A &= 2\pi r^2 + 2\pi r h \\ &= 2\pi r(r+h) \\ &= 2\pi r(r+kr) \\ A &= 2\pi r^2(k+1) \end{aligned}$$

$$\frac{dA}{dr} = 4\pi(k+1)r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

when $r=4$, $\frac{dA}{dt} = 0.1$

$$\frac{1}{10} = 16\pi(k+1) \times \frac{1}{240k\pi}$$

$$\frac{3}{2} = \frac{k+1}{k}$$

$$3k = 2k+2$$

$$k = 2$$

③ correct solution

② obtains correct equation in terms of k

① finds $\frac{dA}{dr}$

14b) i) P H O T G R A S
P H O

$$\text{Arrangements} = \frac{11!}{2!2!2!} = 4989600$$

② correct solution
① calculates 11!

bii)

H _ _ _ _ H _ _ _ _

Second H can go in 6 positions

Remaining letters can be arranged in $\frac{9!}{2!2!}$ ways

$$\text{Total arrangements} = 6 \times \frac{9!}{2!2!}$$

$$= 544320$$

$$\therefore \text{Probability} = \frac{544320}{4989600} = \frac{6}{55}$$

② correct solution
① considers possible positions of H's
or ① determines ways of arranging other 9 letters
(must have explained their calculations)

$$\begin{aligned} c) {}^{n+1}C_r - {}^nC_r &= \frac{(n+1)!}{(n+1-r)!r!} - \frac{n!}{(n-r)!r!} \\ &= \frac{(n+1)!}{(n+1-r)!r!} - \frac{(n-r+1)n!}{(n-r+1)!r!} \\ &= \frac{(n+1)n! - (n-r+1)n!}{(n-r+1)!r!} \\ &= \frac{n!(n+1 - (n-r+1))}{(n-r+1)!r!} \\ &= \frac{n!r}{(n-r+1)!r(r-1)!} \end{aligned}$$

② correct solution
① obtains a common denominator

$$= \frac{n!}{(n-r+1)!(r-1)!}$$

$$= \frac{n!}{(n-(r-1))!(r-1)!}$$

$$= {}^n C_{r-1}$$

$$\text{ii)} \quad {}^3 C_2 + {}^4 C_2 + {}^5 C_2 + \dots + {}^{100} C_2$$

$$= ({}^4 C_3 - {}^3 C_3) + ({}^5 C_3 - {}^4 C_3) + ({}^6 C_3 - {}^5 C_3) + \dots + ({}^{101} C_3 - {}^{100} C_3)$$

$$= {}^{101} C_3 - {}^3 C_3$$

$$= 166649$$

② correct solution

① uses identity in (i)